

Master Thesis
BlueRov2 Formation Control

Week 1

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Initial Plan for Upcoming Weeks

1. Formation control with single integrator agents no constraints
    - Undirected graph
    - Directed graph

2. Single integrator formation control with constraints
    - Only connectivity maintenance
    - Then also colision avoidance

3. Double integrator agents

4. BlueRov2 model no input constraints

5. [Research challenge] integrate input constraints

6. [Research challenge] controller robust or less sensitive to uncertainties (e.g. inverse optimal controller in the sense of LgV like control laws rather than feedback linearization)

Achievements

- Study
  - Slotine’s videos on nonlinear control
  - Esteban’s thesis

- Implementation
  - Formation control with single integrator agents and no constraints
    - Undirected graph (2D and 3D)
    - Directed graph (3D)

- Others
  - GitHub repository

Implementation - Formation control in an undirected system

Initial Information

3 robots/agents

Initial Pose

Final/Desired Pose

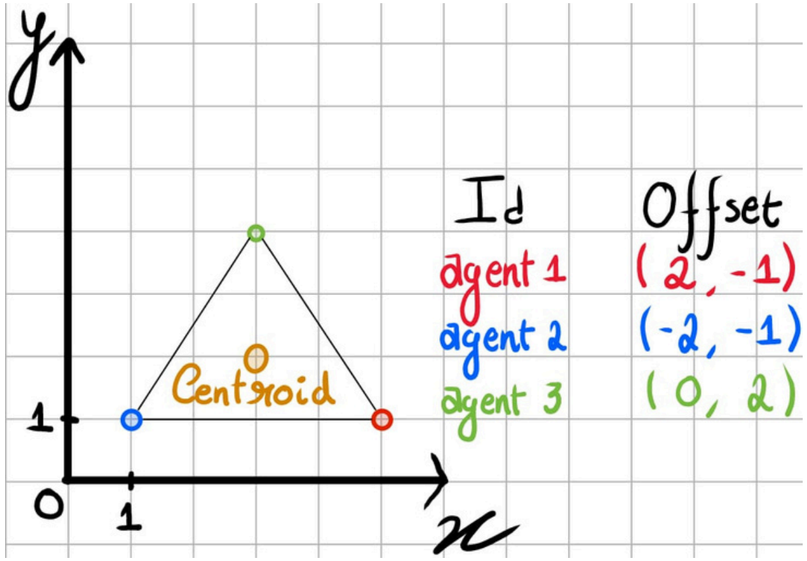
Needed

Agent's velocity throughout time

Goal

Triangle Formation

Offset relative to triangle centroid



How

$x_i \in \mathbb{R}^n$ is the state of each agent

$$\dot{x}_i = u_i, \quad i \leq N,$$

$$A = [a_{ij}] \in \mathbb{R}^{N \times N}$$

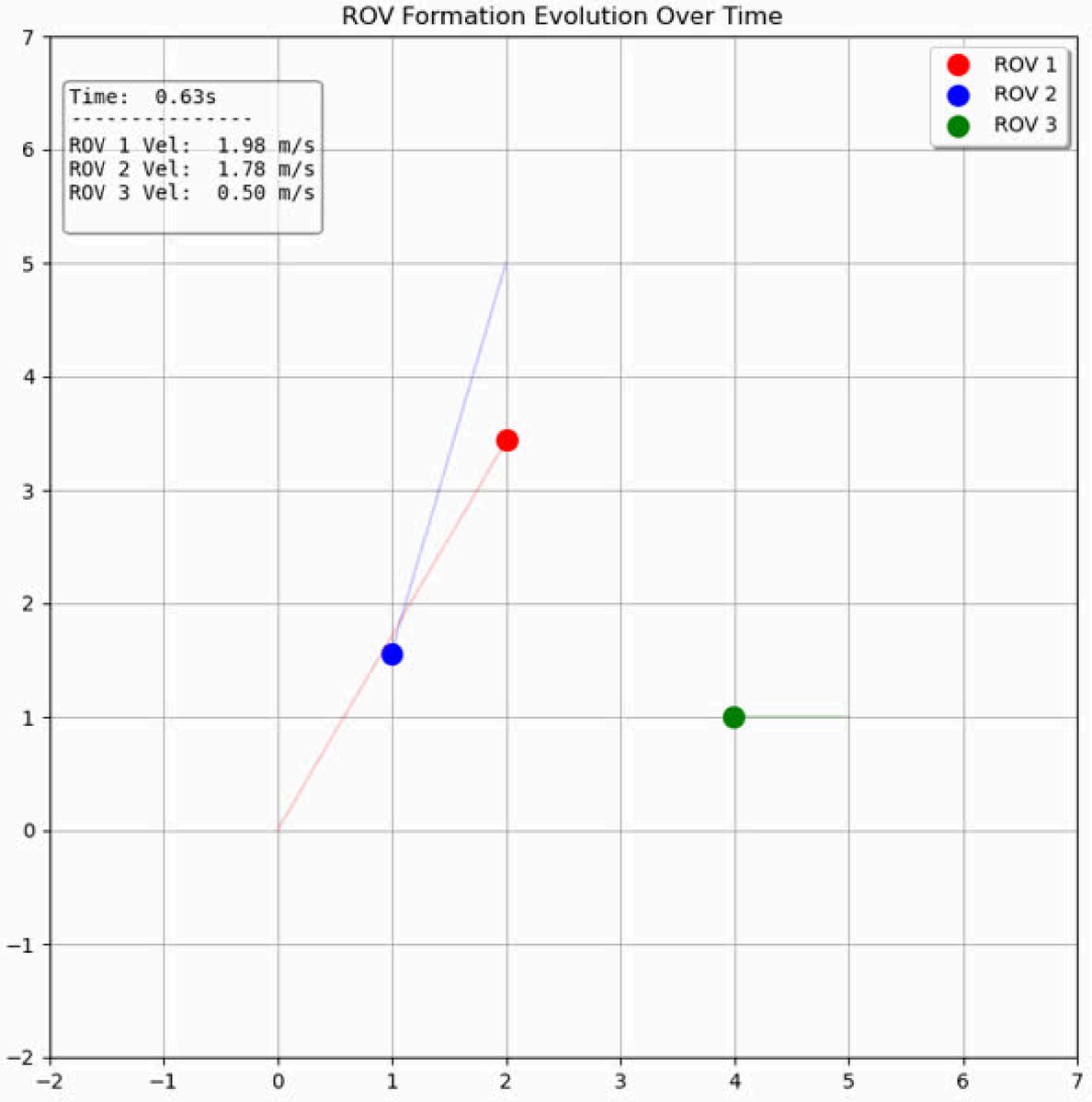
$a_{ij} > 0$ if and only if $(j, i) \in \mathcal{E}$ and $a_{ij} = 0$ otherwise.

$$u_i = -c \sum_{j=1}^{N} a_{ij}(x_i - x_j), \quad c > 0,$$

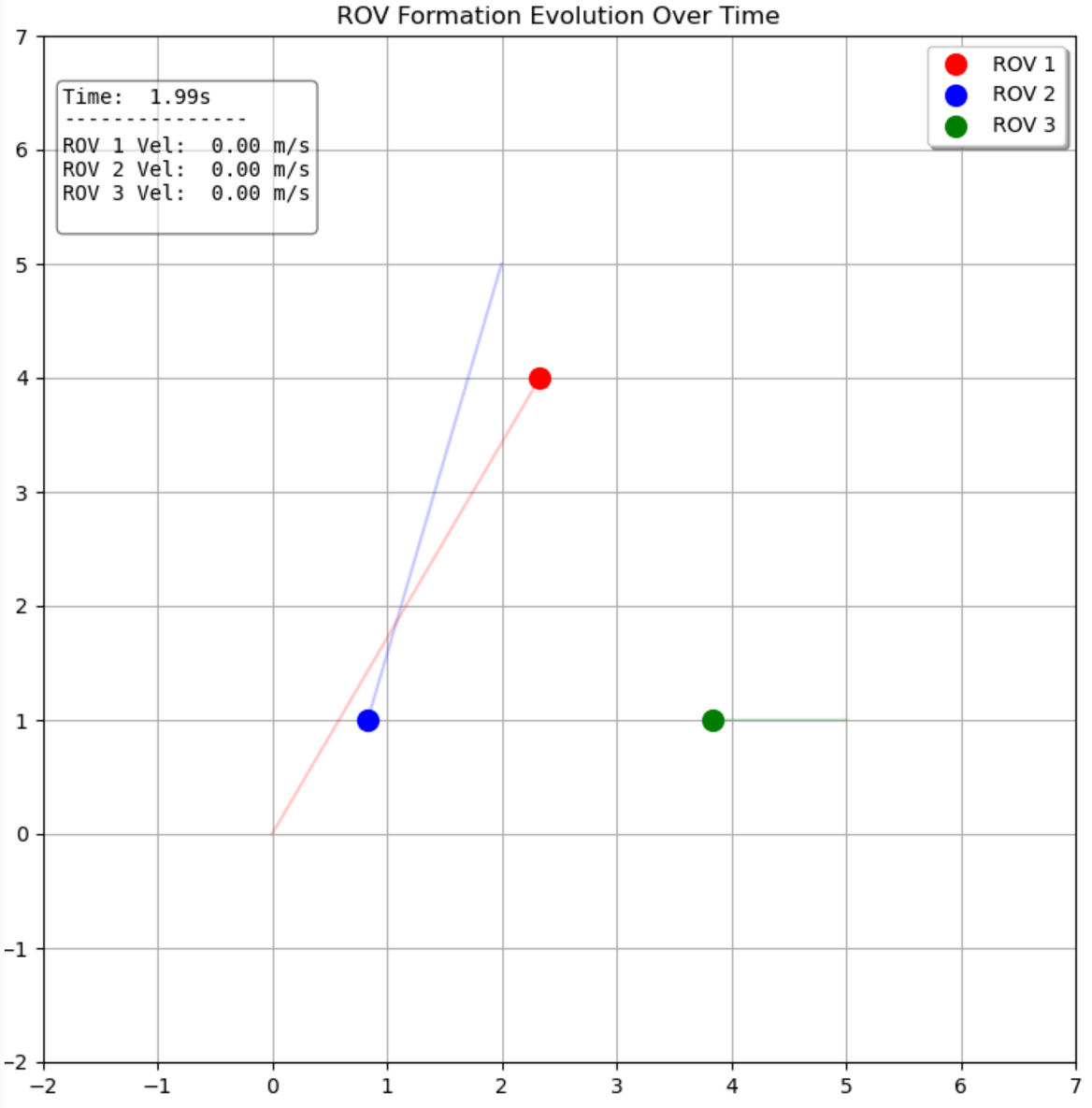
$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$u_i = -\sum_{j \in \mathcal{N}_i} [(x_i - x_j) - (o_i - o_j)]$$

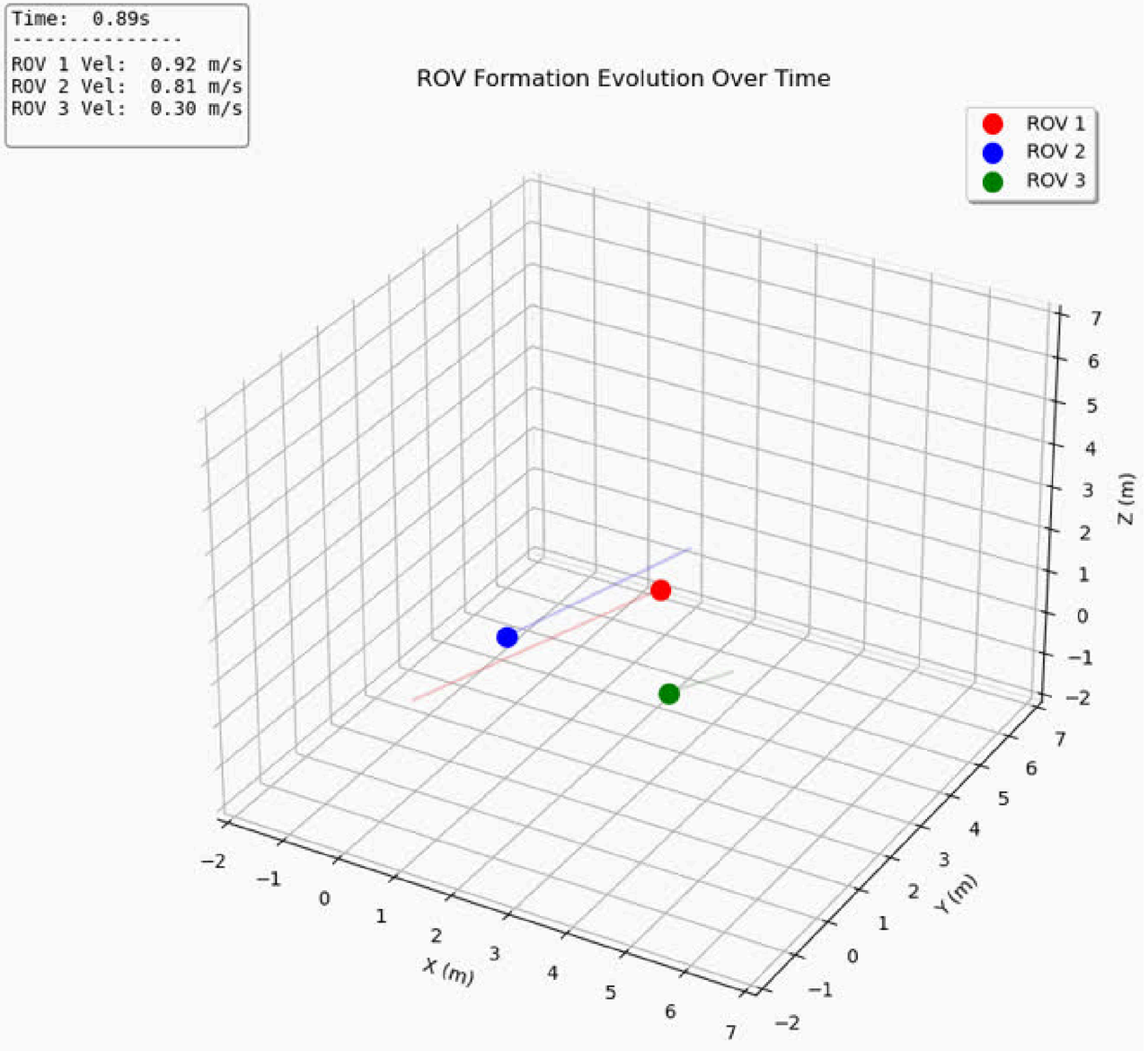
$$x_{i+1} = x_i + u_i \Delta t$$


ROV Formation Evolution Over Time
Time: 0.63s

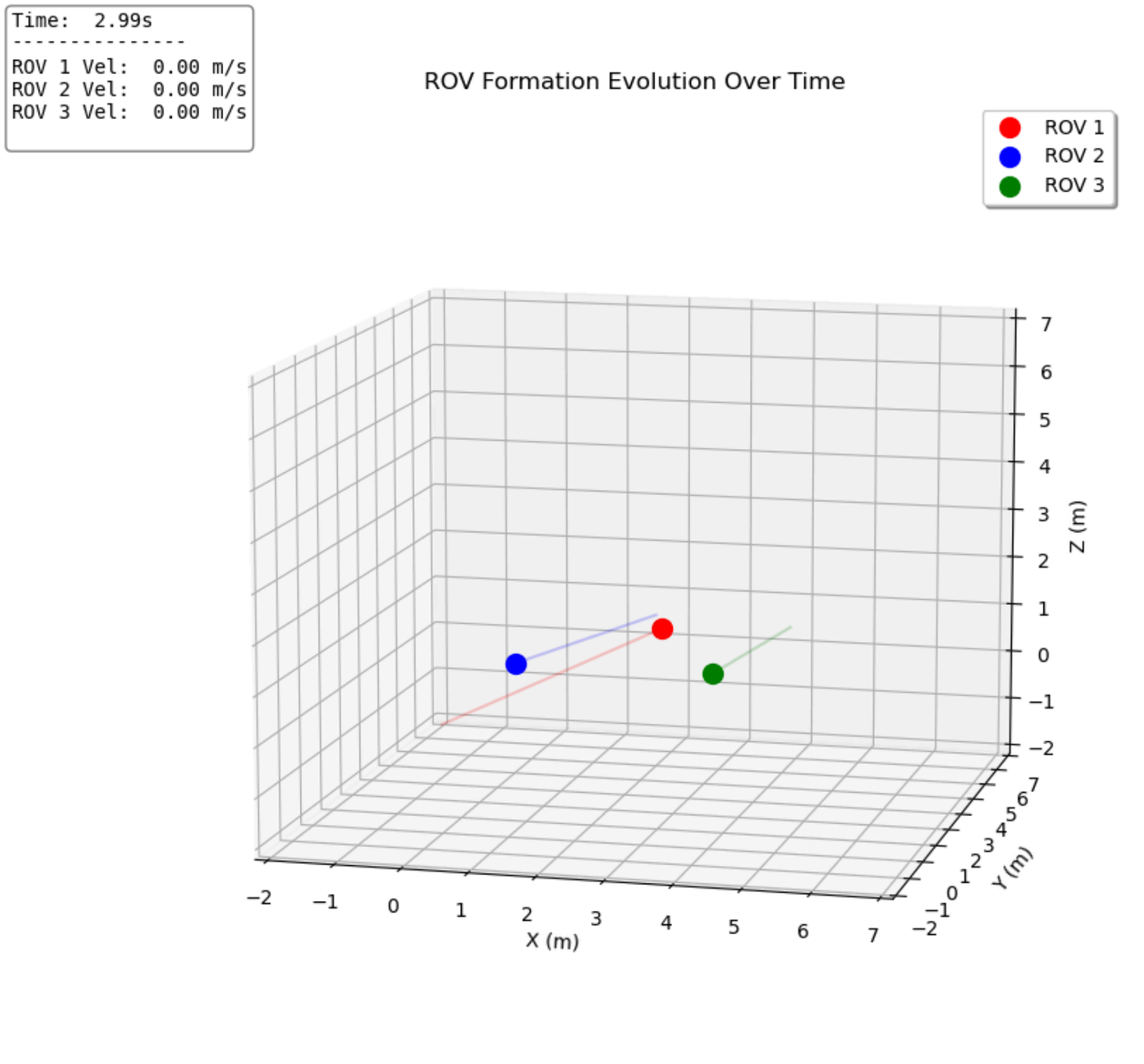
ROV 1 Vel: 1.98 m/s
ROV 2 Vel: 1.78 m/s
ROV 3 Vel: 0.50 m/s
ROV 1
ROV 2
ROV 3


ROV Formation Evolution Over Time
Time: 1.99s

ROV 1 Vel: 0.00 m/s
ROV 2 Vel: 0.00 m/s
ROV 3 Vel: 0.00 m/s
ROV 1
ROV 2
ROV 3


Time: 0.89s

ROV 1 Vel: 0.92 m/s
ROV 2 Vel: 0.81 m/s
ROV 3 Vel: 0.30 m/s
ROV Formation Evolution Over Time
ROV 1
ROV 2
ROV 3
X (m)
Y (m)
Z (m)


Time: 2.99s

ROV 1 Vel: 0.00 m/s
ROV 2 Vel: 0.00 m/s
ROV 3 Vel: 0.00 m/s
ROV Formation Evolution Over Time
ROV 1
ROV 2
ROV 3
X (m)
Y (m)
Z (m)

Implementation - Formation control in a directed system

Initial Information

3 robots/agents

Initial Pose

Leader's velocity/trajectory over time

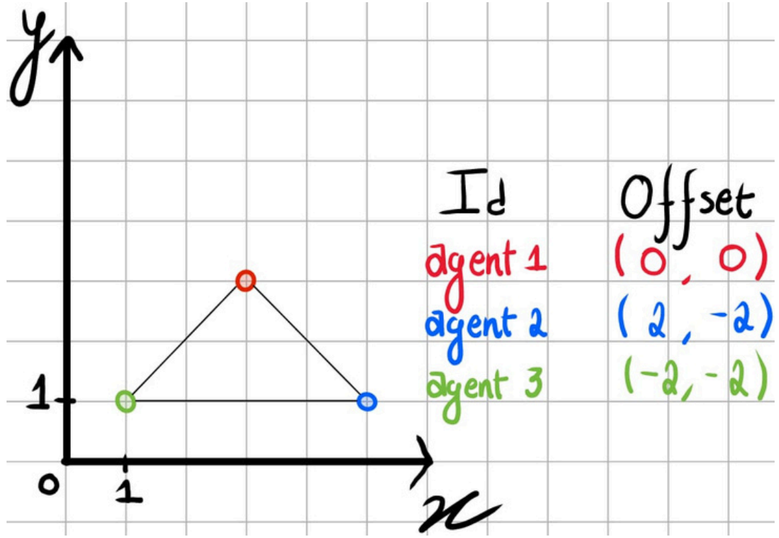
Needed

Other agent's velocity throughout time

Goal

Triangle Formation

Offset relative to leader



How

$x_i \in \mathbb{R}^n$ is the state of each agent

$$\dot{x}_i = u_i, \quad i \leq N,$$

$$u_i = -c \sum_{j=1}^{N} a_{ij}(x_i - x_j), \quad c > 0,$$

$$u_i = -\sum_{j \in \mathcal{N}_i} [(x_i - x_j) - (o_i - o_j)]$$

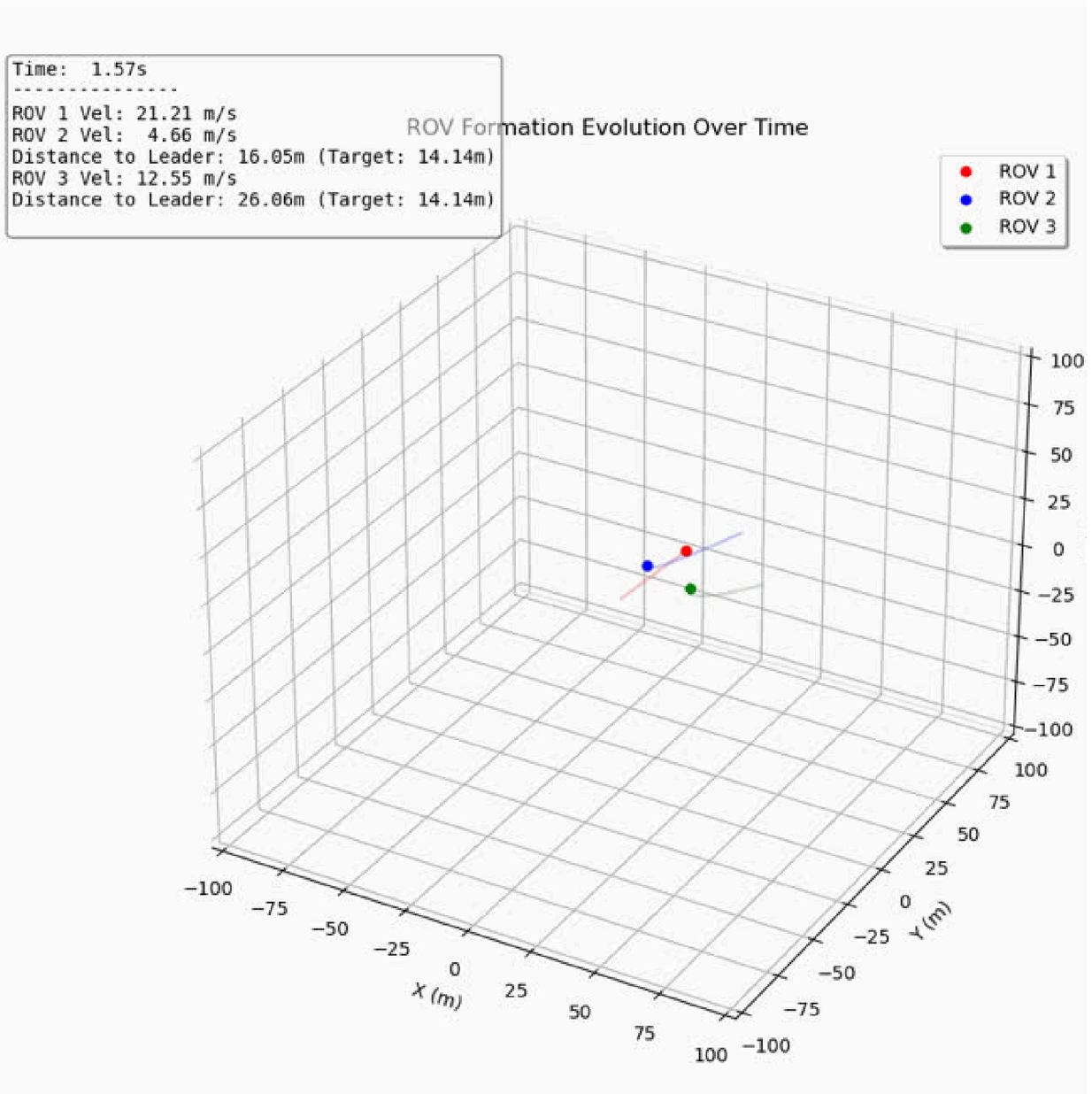
$$A = [a_{ij}] \in \mathbb{R}^{N \times N}$$

$a_{ij} > 0$ if and only if $(j, i) \in \mathcal{E}$ and $a_{ij} = 0$ otherwise.

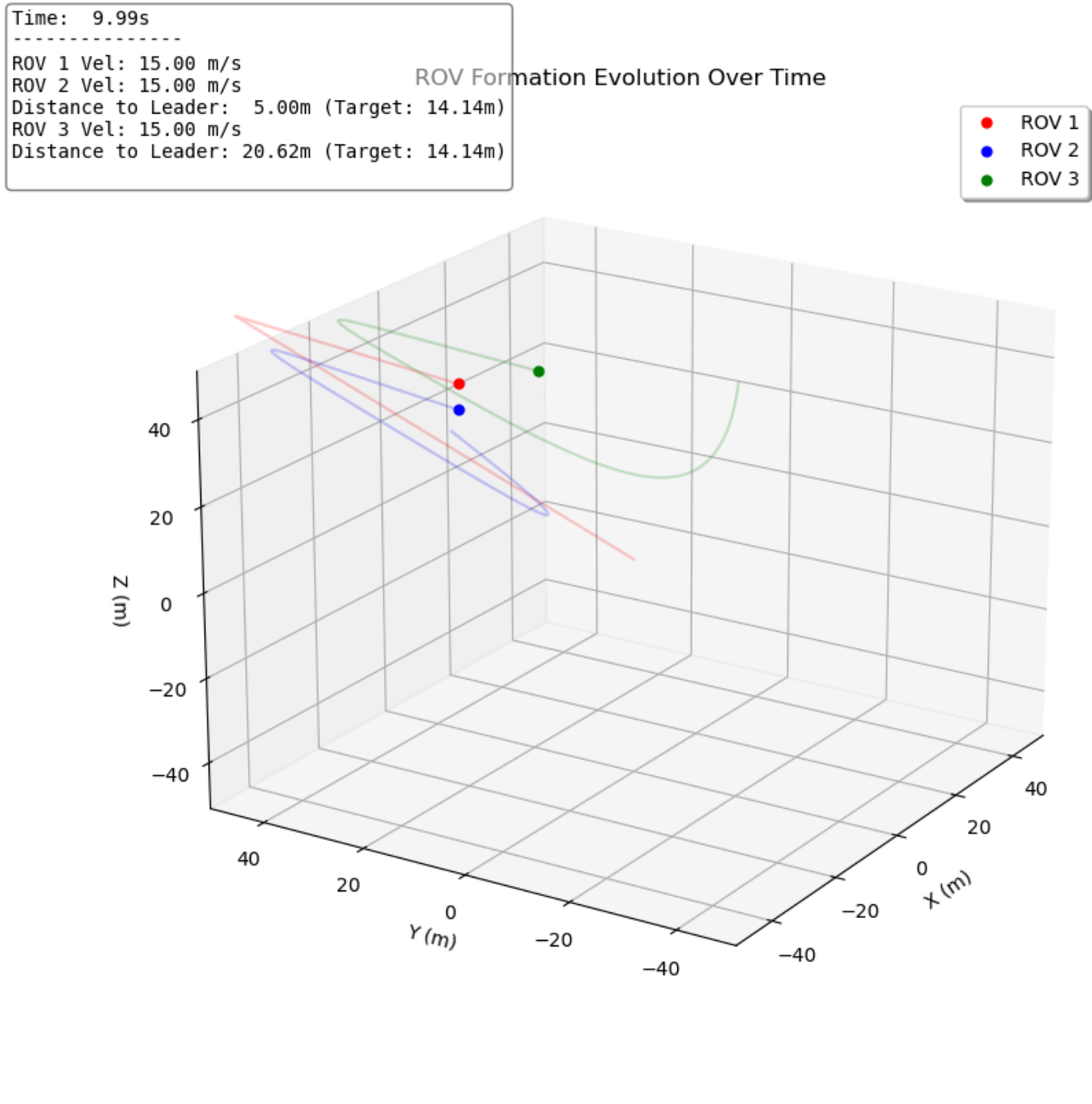
$$A = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$x_{i+1} = x_i + u_i \Delta t$$

Issues


Time: 1.57s

ROV 1 Vel: 21.21 m/s
ROV 2 Vel: 4.66 m/s
Distance to Leader: 16.05m (Target: 14.14m)
ROV 3 Vel: 12.55 m/s
Distance to Leader: 26.06m (Target: 14.14m)
ROV Formation Evolution Over Time
ROV 1
ROV 2
ROV 3
X (m)
Y (m)


Time: 9.99s

ROV 1 Vel: 15.00 m/s
ROV 2 Vel: 15.00 m/s
Distance to Leader: 5.00m (Target: 14.14m)
ROV 3 Vel: 15.00 m/s
Distance to Leader: 20.62m (Target: 14.14m)
ROV Formation Evolution Over Time
ROV 1
ROV 2
ROV 3
Z (m)
X (m)
Y (m)

Solution

Formation Offset

Fixed Point

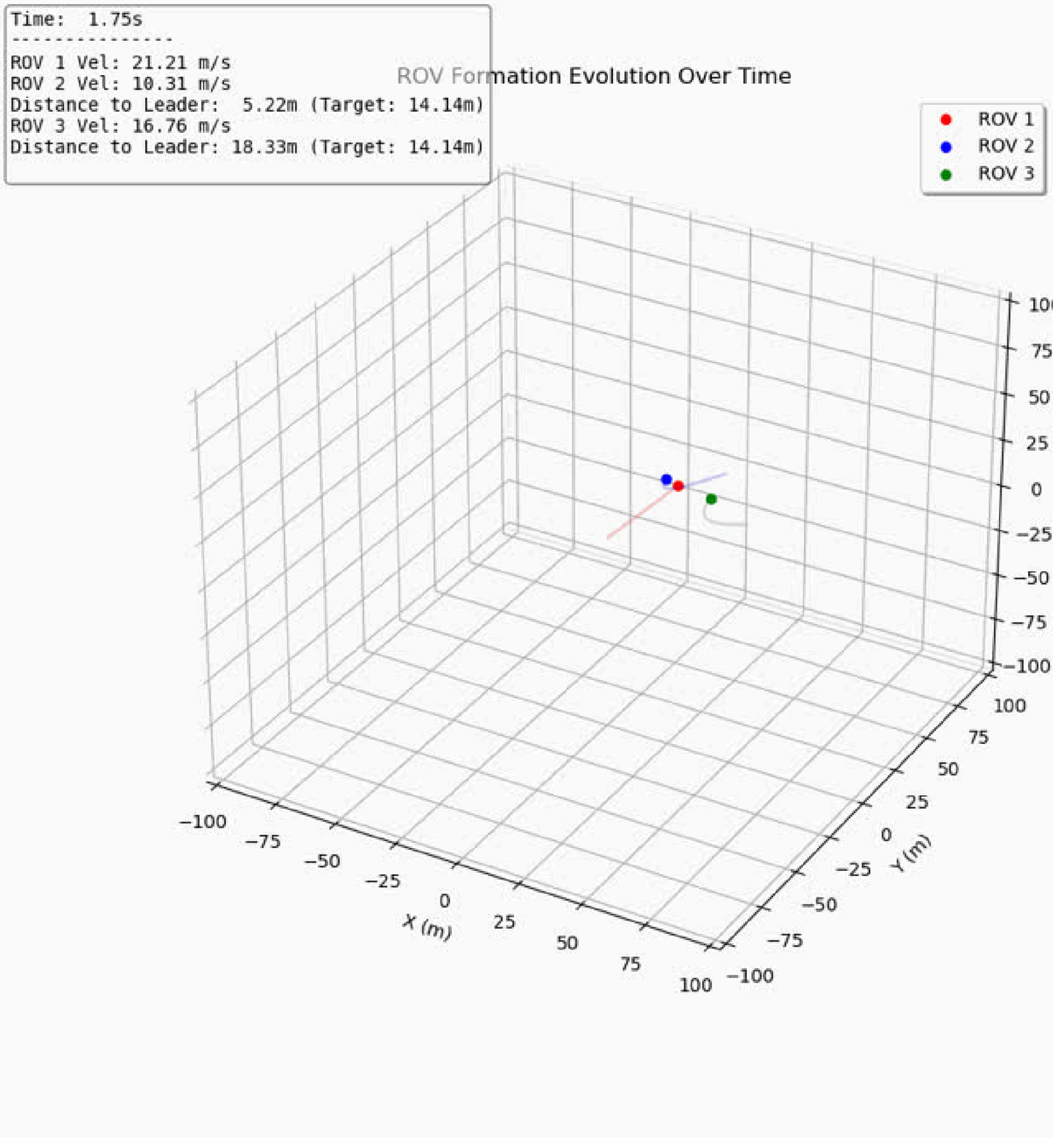
$$u_i = -\sum_{j \in \mathcal{N}_i} [(x_i - x_j) - (o_i - o_j)]$$

VS

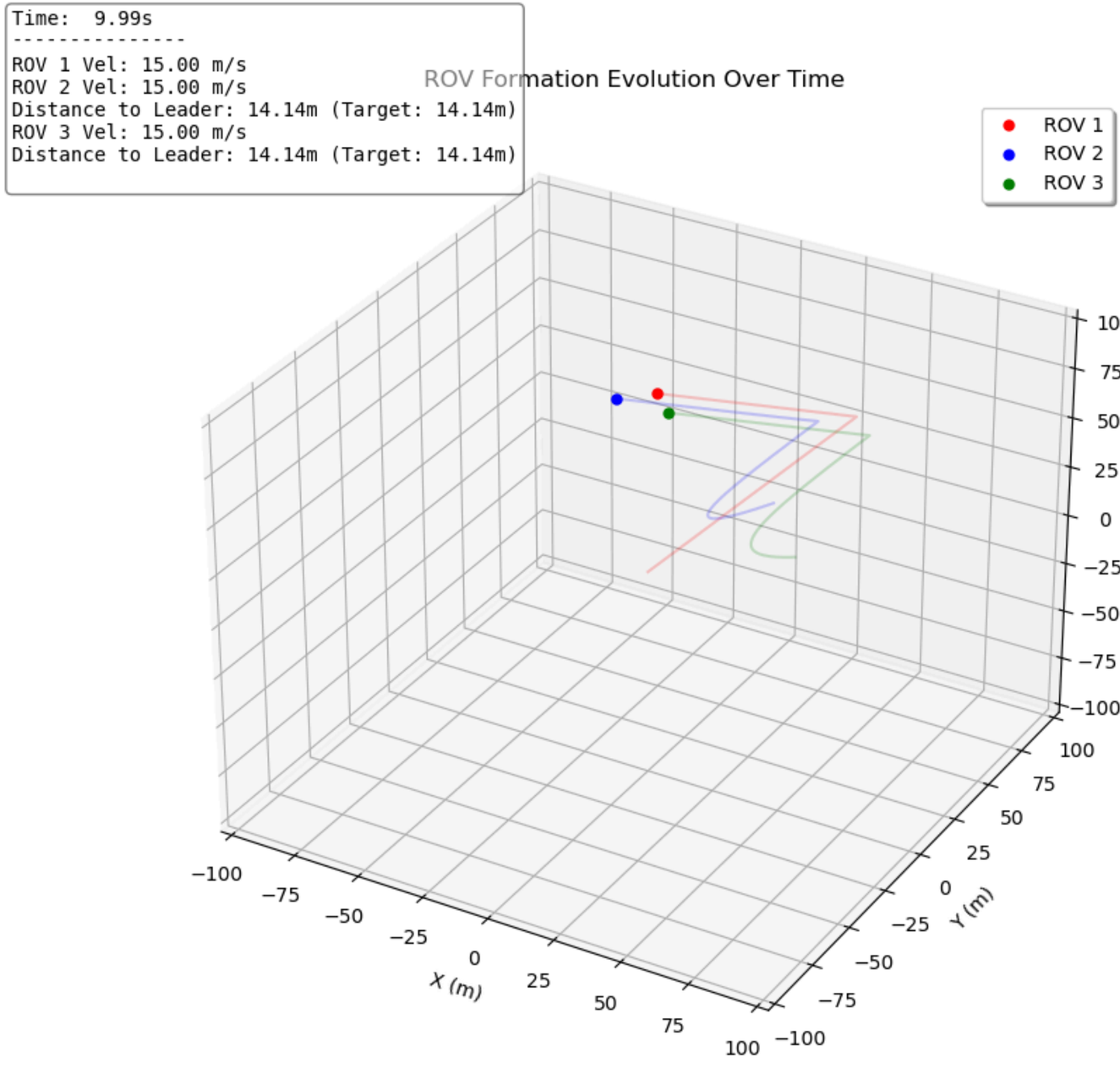
Moving/Dynamic Point

$$u_i = -(x_i - x_j) - (o_i - o_j) + u_j\,, \quad j = Leader's\ \ Id$$

Leader's movement must be taken into account


Time: 1.75s

ROV 1 Vel: 21.21 m/s
ROV 2 Vel: 10.31 m/s
Distance to Leader: 5.22m (Target: 14.14m)
ROV 3 Vel: 16.76 m/s
Distance to Leader: 18.33m (Target: 14.14m)
ROV Formation Evolution Over Time
ROV 1
ROV 2
ROV 3
X (m)
Y (m)


Time: 9.99s

ROV 1 Vel: 15.00 m/s
ROV 2 Vel: 15.00 m/s
Distance to Leader: 14.14m (Target: 14.14m)
ROV 3 Vel: 15.00 m/s
Distance to Leader: 14.14m (Target: 14.14m)
ROV Formation Evolution Over Time
ROV 1
ROV 2
ROV 3
X (m)
Y (m)

Next Steps

- Formation control with single integrator agents no constraints
  - Undirected graph
  - Directed graph
- Single integrator formation control with constraints
  - Only connectivity maintenance
  - Then also colision avoidance
- Double integrator agents
- BlueRov2 model no input constraints
- [Research challenge] integrate input constraints
- [Research challenge] controller robust or less sensitive to uncertainties (e.g. inverse optimal controller in the sense of LgV like control laws rather than feedback linearization)



Questions & Comments